

ALFAAL00975

## Lawesson's Reagent

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                                       |                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b>           | <b>Lawesson's Reagent</b><br><b>Lawesson's Reagent</b>                                                                                                                                                                                                             |
| <b>Cat No. :</b>                                      | <b>L00975</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                                       | 2,4-Bis(4-methoxyphenyl)-1,3-dithia-2,4-diphosphetane-2,4-disulfide                                                                                                                                                                                                |
| <b>CAS No</b>                                         | 19172-47-5                                                                                                                                                                                                                                                         |
| <b>Molecular Formula</b>                              | C14 H14 O2 P2 S4                                                                                                                                                                                                                                                   |
| <b>Supplier</b>                                       | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>                     | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                                 | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b><br><b>Uses advised against</b> | Laboratory chemicals.<br>No Information available                                                                                                                                                                                                                  |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                |                                   |                       |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-----------------------|
| <b>Physical State</b><br>Powder Solid                                                                                                          | <b>Appearance</b><br>Light yellow | <b>Odor</b><br>Stench |
| <b>Emergency Overview</b><br>In contact with water releases flammable gas. Contact with water liberates toxic gas. Moisture sensitive. Stench. |                                   |                       |

#### Classification of the substance or mixture

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 2 |
|------------------------------------------------------------------------|------------|

#### Label Elements



**Signal Word**

**Danger**

#### **Hazard Statements**

H261 - In contact with water releases flammable gases

**Precautionary Statements****Prevention**

P231 + P232 - Handle and store contents under inert gas. Protect from moisture

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P402 + P404 - Store in a dry place. Store in a closed container

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

In contact with water releases flammable gas.

**Health Hazards**

The product contains no substances which at their given concentration are considered to be hazardous to health.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Is not likely mobile in the environment. Decomposes in contact with water.

**Other Hazards**

Stench. This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component                                                             | CAS No     | Weight % |
|-----------------------------------------------------------------------|------------|----------|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | <=100    |

**SECTION 4. FIRST AID MEASURES****General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Most important symptoms and effects**

None reasonably foreseeable.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES**

**Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

Contact with water liberates toxic gas.

**Specific Hazards Arising from the Chemical**

Water reactive. Contact with water liberates toxic gas. Produce flammable gases on contact with water. Contact with water liberates toxic gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Do not expose spill to water.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Do not allow contact with water.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep from any possible contact with water. Keep away from water or moist air. Store under an inert atmosphere.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation

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or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                   |                                 |                                          |
|-----------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                 | Light yellow                    |                                          |
| <b>Physical State</b>             | Powder Solid                    |                                          |
| <b>Odor</b>                       | Stench                          |                                          |
| <b>Odor Threshold</b>             | No data available               |                                          |
| <b>pH</b>                         | Not applicable                  |                                          |
| <b>Melting Point/Range</b>        | 228 - 231 °C / 442.4 - 447.8 °F |                                          |
| <b>Softening Point</b>            | No data available               |                                          |
| <b>Boiling Point/Range</b>        | No information available        |                                          |
| <b>Flash Point</b>                | No information available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>           | Not applicable                  | Solid                                    |
| <b>Flammability (solid,gas)</b>   | No information available        |                                          |
| <b>Explosion Limits</b>           | No data available               |                                          |
| <b>Vapor Pressure</b>             | No data available               |                                          |
| <b>Vapor Density</b>              | Not applicable                  | Solid                                    |
| <b>Specific Gravity / Density</b> | No data available               |                                          |
| <b>Bulk Density</b>               | No data available               |                                          |

|                                         |                          |       |
|-----------------------------------------|--------------------------|-------|
| Water Solubility                        | Decomposes               |       |
| Solubility in other solvents            | No information available |       |
| Partition Coefficient (n-octanol/water) |                          |       |
| Autoignition Temperature                | Not applicable           |       |
| Decomposition Temperature               | No data available        |       |
| Viscosity                               | Not applicable           | Solid |
| Explosive Properties                    | No information available |       |
| Oxidizing Properties                    | No information available |       |
|                                         |                          |       |
| Molecular Formula                       | C14 H14 O2 P2 S4         |       |
| Molecular Weight                        | 404.45                   |       |

**SECTION 10. STABILITY AND REACTIVITY**

|                                  |                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|
| Stability                        | Moisture sensitive.                                                                                        |
| Hazardous Reactions              | None under normal processing.                                                                              |
| Hazardous Polymerization         | No information available.                                                                                  |
| Conditions to Avoid              | Incompatible products. Exposure to moist air or water. Exposure to moisture.                               |
| Materials to avoid               | Water. Strong oxidizing agents.                                                                            |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Sulfur oxides. Phosphorus trihydride (phosphine). |

**SECTION 11. TOXICOLOGICAL INFORMATION**

|                                        |                                                             |
|----------------------------------------|-------------------------------------------------------------|
| Product Information                    | No acute toxicity information is available for this product |
| (a) acute toxicity;                    |                                                             |
| (b) skin corrosion/irritation;         | No data available                                           |
| (c) serious eye damage/irritation;     | No data available                                           |
| (d) respiratory or skin sensitization; |                                                             |
| Respiratory                            | No data available                                           |
| Skin                                   | No data available                                           |
| (e) germ cell mutagenicity;            | No data available                                           |
| (f) carcinogenicity;                   | No data available                                           |
|                                        | There are no known carcinogenic chemicals in this product   |
| (g) reproductive toxicity;             | No data available                                           |
| (h) STOT-single exposure;              | No data available                                           |
| (i) STOT-repeated exposure;            | No data available                                           |
| Target Organs                          | None known.                                                 |

(j) aspiration hazard; Not applicable  
Solid

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed** No information available

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** No information available  
**Persistence** Persistence is unlikely, based on information available.  
**Degradability** Decomposes in contact with water.  
**Degradation in sewage treatment plant** No information available. Decomposes in contact with water.

**Bioaccumulative Potential** Product does not bioaccumulate due to reaction with water

**Mobility in soil** Decomposes in contact with water Is not likely mobile in the environment

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14. TRANSPORT INFORMATION

### Road and Rail Transport

**UN-No** UN3134  
**Proper Shipping Name** Water-reactive solid, toxic, n.o.s.  
**Technical Shipping Name** Lawesson's Reagent  
**Hazard Class** 4.3  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

### IMDG/IMO

## SAFETY DATA SHEET

## Lawesson's Reagent

UN-No UN3134  
Proper Shipping Name Water-reactive solid, toxic, n.o.s.  
Technical Shipping Name Lawesson's Reagent  
Hazard Class 4.3  
Subsidiary Hazard Class 6.1  
Packing Group II

IATA

UN-No UN3134  
Proper Shipping Name WATER-REACTIVE SOLID, TOXIC, N.O.S.\*  
Technical Shipping Name Lawesson's Reagent  
Hazard Class 4.3  
Subsidiary Hazard Class 6.1  
Packing Group II

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                                             | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|------|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | -                                                   | -                                       | X    | X     | 242-855-4 | X    | -   | X     | -    | X    | X    | -    |

## National Regulations

## SECTION 16. OTHER INFORMATION

Prepared By Health, Safety and Environmental Department  
Creation Date 29-Sep-2006  
Revision Date 25-Apr-2024  
Revision Summary New emergency telephone response service provider.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Legend

CAS - Chemical Abstracts Service

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

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**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances  
**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**DNEL** - Derived No Effect Level  
**RPE** - Respiratory Protective Equipment  
**LC50** - Lethal Concentration 50%  
**NOEC** - No Observed Effect Concentration  
**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**PNEC** - Predicted No Effect Concentration  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road  
**OECD** - Organisation for Economic Co-operation and Development  
**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**ATE** - Acute Toxicity Estimate  
**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**